

Research Statement

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My research sits at the intersection of network science, measurement, causal inference, Bayesian statistics, and deep learning. I develop statistical and causal models for data on people who are connected to one another, with particular attention to settings where relationships, norms, and incentives shape outcomes, and where an intervention changes not only behavior but also the social structure that sustains it. I am motivated by questions that can be answered with real data, where careful modeling yields both empirical insight and theoretical clarity.

Field studies of interventions in social networks

At Yale's Human Nature Lab, working with Professor Nicholas Christakis, Professor Laura Forastiere, and Professor Feng Fu, I analyze sociocentric village networks in rural Honduras to study how social structure shapes behavior and how a large-scale intervention reshapes that structure.

The central result of this line of work is *Educational Intervention Can Rewire Village Social Networks*, accepted at *Nature* with Laura Forastiere, Edoardo M. Airolidi, and Nicholas A. Christakis. Using a two-stage randomized public-health intervention in 110 remote villages covering 8,331 individuals, we quantify how an informational intervention alters pre-existing sociocentric networks. In low-dosage villages, treated individuals tend to sever health-advice ties with untreated alters. In high-dosage villages, treated individuals increase both inbound and outbound ties. The intervention also moves friendship and financial ties, which indicates multiplex spillovers, while aggregate village-level structure changes only modestly. The substantive implication matters for anyone who evaluates interventions in connected populations. An intervention can change the medium through which its own effect propagates, so a design that treats the network as fixed will misstate what the intervention did.

A second line studies how social and individual characteristics support cooperation. In repeated public-goods experiments across 134 villages with 2,591 participants, contribution trajectories split early into two persistent regimes, low and high, with little subsequent convergence. Groups with an early majority of above-norm contributors are much more likely to settle into the high-cooperation regime, and early high contributions by socially central individuals help move groups onto that path. A third line, in collaboration with the team at the Human Nature Lab, examines how village social structure relates to depression, generalized anxiety, and attitudes supportive of domestic violence. The depression study appeared in *BMJ Open*, the anxiety study is under review there, and further work on the spatial structure of friendly and antagonistic ties and on static and dynamic tie formation is under review at the *Journal of Complex Networks*.

Measurement and measurement error in relational data

Relational data are routinely analyzed as if observed, but they are reported. Respondents forget ties, name people who do not name them back, and answer differently depending on how the instrument is worded and how long the roster is. In *Forgetting Friends and Foes: Self-Reported Errors in Sociocentrically Mapped Village Networks*, my coauthors and I quantify systematic reporting error in a large sociocentric census and characterize which downstream conclusions survive it.

This motivates a modeling response. In *Clustering Network Formation in Populations of Networks*, under revision at the *Annals of Applied Statistics*, we treat each observed network as a noisy snapshot of an underlying network, and we cluster networks jointly by their strategic formation mechanism and by their measurement-error profile. The likelihood is built from a single snapshot using a meeting-and-revision equilibrium, with a finite-horizon alternative, and inference is a parallelizable block-updated MCMC scheme with exchange-type updates. Putting the reporting process into the model, as a parameter to be estimated instead of an assumption to be hoped for, is what allows the formation mechanism to be identified at all.

Robustness, sensitivity, and what a conclusion can survive

A defining theme of my work is methodological clarity in settings where network-based conclusions are sensitive to modeling choices. With Regina Ruane, Swati Chandna, and Simón Lunagómez, in *Decision-Theoretic Robustness for Network Models*, under revision at the *Journal of Machine Learning Research*, we develop a decision-theoretic notion of robustness by allowing the posterior to vary within a small Kullback–Leibler neighborhood. For exchangeable graphs we connect robust posterior risk to graphon limits, which yields tractable sensitivity expansions for losses defined on network functionals, and we give a practical computation via exponential tilting of posterior draws. The output is a diagnostic an applied researcher can read directly. It states how far a conclusion travels before a small and plausible departure from the fitted model reverses it, which separates substantive findings from artifacts of modeling convenience.

Latent variable models, model combination, and geometry

With Professor Edoardo Airoldi and Professor Simón Lunagómez I study latent space network models with hyperbolic and spherical geometries, addressing identifiability by characterizing which latent representations are recoverable from observed links and by imposing constraints that resolve the non-identifiability that remains. In related work we introduce a general framework for wrapped distributions on homogeneous Riemannian manifolds, built from area-preserving maps and isometries, to provide tractable priors for variational autoencoders and latent network models. With Professor Regina Ruane I develop radial compensation distributions, which stabilize manifold-based generative models by decoupling curvature from parameter semantics, yielding closed-form, invariant, and less distorted distributions in high-dimensional Riemannian spaces.

A closely related question is what to do when several plausible models compete. In *Bayesian Predictive Synthesis for Dynamic Networks: Forecasting and Identifying Structural Mechanisms*, with Regina Ruane and Theofanis Papamichalis, under second-round revise and resubmit at the *Journal of the American Statistical Association*, we combine competing generative models for dynamic networks at the level of the graphon. Forecasts become calibrated, and the weights assigned to component models are interpretable as evidence for competing structural mechanisms of tie formation and dissolution. Model combination becomes a way of asking which mechanism the data support, and not only a way of improving prediction.

Most recently, *Support-Safe Variational Hybrid Filtering for Contact-Mode and Sparse-Law Recovery*, with Regina Ruane, was accepted at NeurIPS. It develops variational filtering for hybrid systems where the active regime and a sparse governing law must be recovered together, under constraints that keep the inferred support well posed.

Emerging direction: decentralized financial networks

In collaboration with Professor Leandros Tassioulas and his team at Yale, I am building a research direction on Ethereum and decentralized finance networks. Using on-chain data, I study how these networks evolve, what mechanisms drive change, and which external events, including macroeconomic announcements, geopolitical shocks, and protocol incidents, coincide with structural breaks. The goal is comparative. I ask how network properties evolve across asset transfers, lending and borrowing, derivatives, and decentralized exchanges, and whether these systems exhibit systematically different patterns of connectivity, concentration, and fragility across market regimes. This identifies which network signatures are common across regimes and which are protocol-specific, which in turn informs how systemic risk propagates through different parts of the ecosystem.

Agenda for the next five years

Three directions organize the program I intend to build.

Design and measurement for relational data in applied settings. I will extend the measurement-error work into design guidance that applied researchers can use before data collection, covering roster and nomination instruments, boundary specification, missing and non-reciprocated ties, and power and sample size calculations for studies that intend to measure or intervene on a network. Studies in schools, classrooms, and organizations collect exactly this kind of data, and the design literature is thinner than the modeling literature.

Causal inference under interference and network change. The *Nature* result implies that the network is itself an outcome. I will develop estimands and designs for settings where treatment changes the interference structure, including strategically targeted rollouts, and sensitivity analyses for the exposure mappings that identification arguments in this area typically assume.

Robust Bayesian latent variable methods, with software. I will extend the decision-theoretic robustness framework beyond exchangeable graphs to multilevel and longitudinal relational data, and I will release documented open-source implementations in R and Python. Methods that stay in papers do not change practice.

I intend to support this program through external funding, with IES, NSF, and NIH as the natural sources, and to continue and expand collaborations with Professor Nicholas Christakis (Yale University), Professor Laura Forastiere (Yale University), Professor Edoardo Airoldi (Temple University), Professor Patrick Wolfe (Purdue University), Professor Leandros Tassioulas (Yale University), Professor Feng Fu (Dartmouth College), Professor Margaret Traeger (University of Notre Dame), Professor Simón Lunagómez (ITAM), and Professor Regina Ruane (University of Pennsylvania). Across domains my core questions are consistent: how networks transmit, mediate, and transform interventions and incentives, how to make reliable inference when relational data are measured with error, and how to design systems and studies that remain informative when the assumptions behind them are only approximately true.